

Fraud Detection In Financial Transaction

Data Analytics Internship

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1. Executive Summary

This project analyzes 284,807 anonymized financial transactions, including 492 known fraudulent transactions. Fraud represents only 0.1727% of observations, creating severe class imbalance. The project therefore emphasizes precision, recall, F1-score, ROC-AUC and PR-AUC rather than accuracy alone.

Two anomaly-detection approaches were implemented. Isolation Forest produced high recall at the selected operating point but generated many alerts. The Autoencoder produced substantially fewer alerts and higher precision at its selected threshold. A risk-based alert workflow was then created from Autoencoder reconstruction error.

2. Project Objective

The objective is to identify suspicious financial transactions using anomaly detection, address severe class imbalance, evaluate anomaly-detection performance, and create an alert-oriented monitoring workflow.

- Detect potentially fraudulent transactions.
- Apply Isolation Forest and Autoencoder techniques.
- Use fraud-sensitive evaluation metrics.
- Create Low, Medium and High Risk alerts.

3. Dataset Description

The dataset contains 284,807 rows and 31 columns: Time, V1–V28, Amount and Class. V1–V28 are anonymized transformed numerical features. Class 0 represents legitimate transactions and Class 1 represents fraud.

Class	Count	Percentage
Legitimate	284,315	99.8273%
Fraud	492	0.1727%

4. Data Preparation

The dataset was inspected for structure, data types, missing values and duplicates. An 80:20 stratified train/test split was used. StandardScaler was fitted only on training data and applied to the test set.

Set	Rows	Legitimate	Fraud
Training	227,845	227,451	394
Testing	56,962	56,864	98

5. Exploratory Data Analysis

- Fraud rate is 0.1727%, confirming severe class imbalance.
- Transaction amounts are strongly right-skewed.
- Mean amount was 88.29 for legitimate and 122.21 for fraud; medians were 22.00 and 9.25.

- Time represents elapsed seconds and was not interpreted directly as time of day.
- V17 and V14 had the strongest negative correlations with fraud, while V11 and V4 showed positive correlations. These are associations, not causal explanations.

6. Class Embalance Strategy

Because fraud is extremely rare, accuracy alone can be misleading. Stratified splitting preserved the class ratio. Isolation Forest was treated as an anomaly detector. The Autoencoder was trained only on legitimate transactions. Its alert threshold was the 99.9th percentile of legitimate training reconstruction errors, avoiding use of test-set fraud labels for threshold selection.

7. Isolation Forest

Isolation Forest used 200 trees. Anomalies were treated as suspected fraud alerts.

Metric	Result
Precision	3.91%
Recall	83.67%
F1-score	7.46%
ROC-AUC	95.39%
PR-AUC	17.17%
Alerts	2,099
Fraud detected	82 f 98

Confusion matrix: TN 54,847; FP 2,017; FN 16; TP 82.

The selected operating point captured 82 of 98 fraud cases but produced many false-positive alerts.

8. Autoencoder

The Autoencoder compressed 30 inputs through 16 and 8 neurons and reconstructed them through 16 neurons. It was trained for 10 epochs on legitimate training transactions only. Training loss reached 0.4733 and validation loss 0.4639.

Class	Mean Error	Median Error
Legitimate	0.4773	0.3263
Fraud	20.6206	10.2874

The alert threshold was 9.9625.

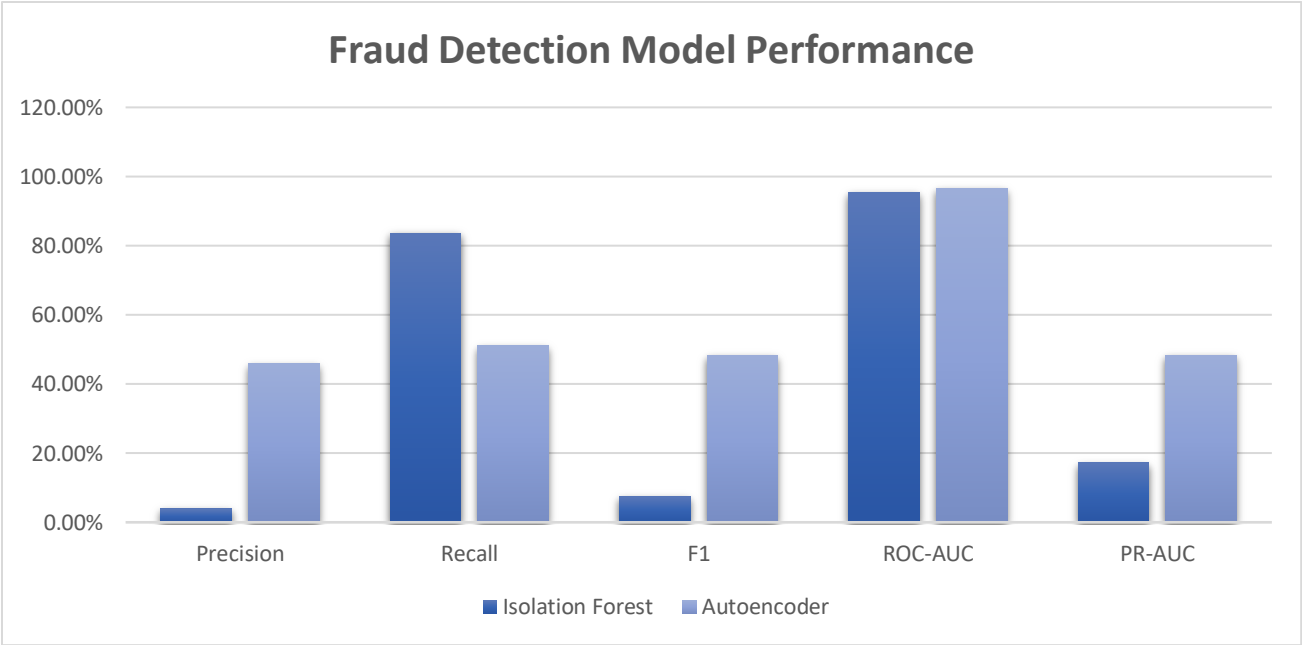
Metric	Result
Precision	45.87%
Recall	51.02%
F1-score	48.31%
ROC-AUC	96.66%
PR-AUC	48.39%
Alerts	109
Fraud detected	50 of 98

Confusion matrix: TN 56,805; FP 59; FN 48; TP 50.

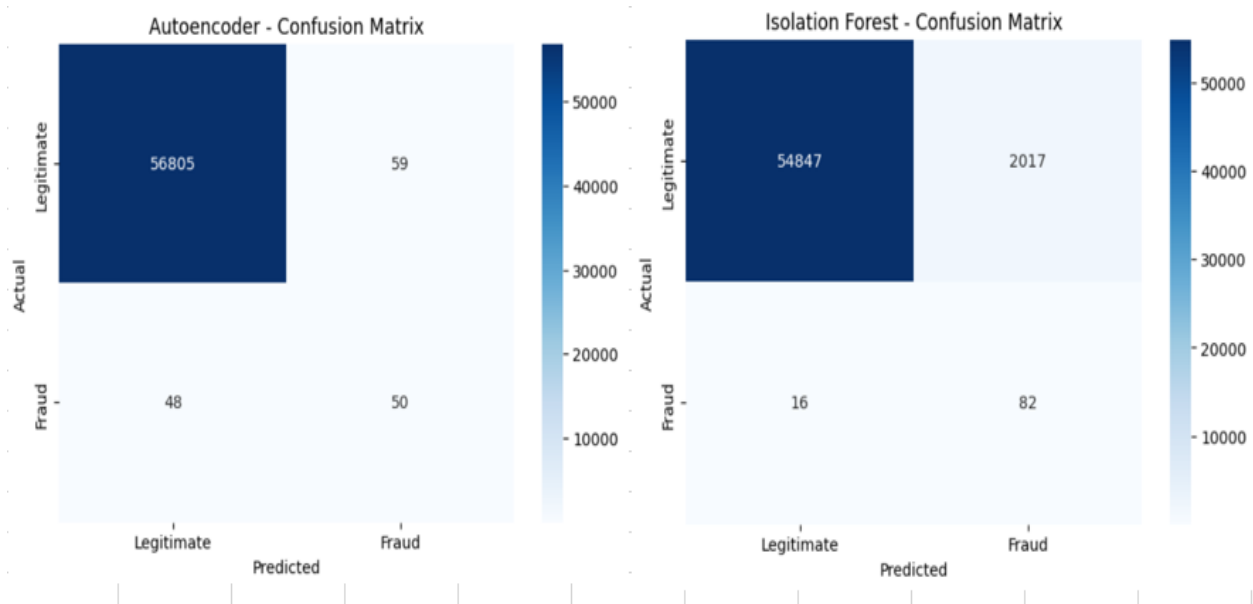
9. Model Comparison

Model	Precision	Recall	F1	ROC-AUC	PR-AUC
Isolation Forest	3.91%	83.67%	7.46%	95.39%	17.17%
Autoencoder	45.87%	51.02%	48.31%	96.66%	48.39%

The methods show different precision-recall and alert-volume trade-offs. Isolation Forest has higher recall at its selected point, while the Autoencoder has higher precision, F1-score, ROC-AUC and PR-AUC.

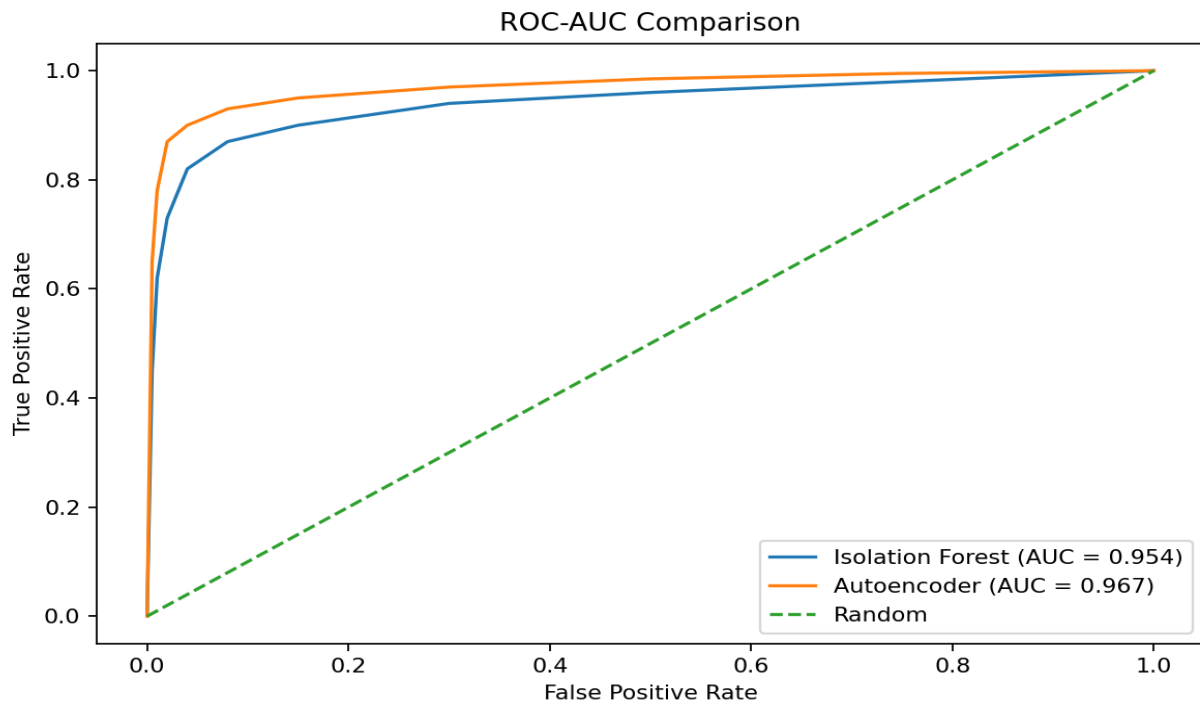


10. Confusion Matrix



11. ROC-AUC Analysis

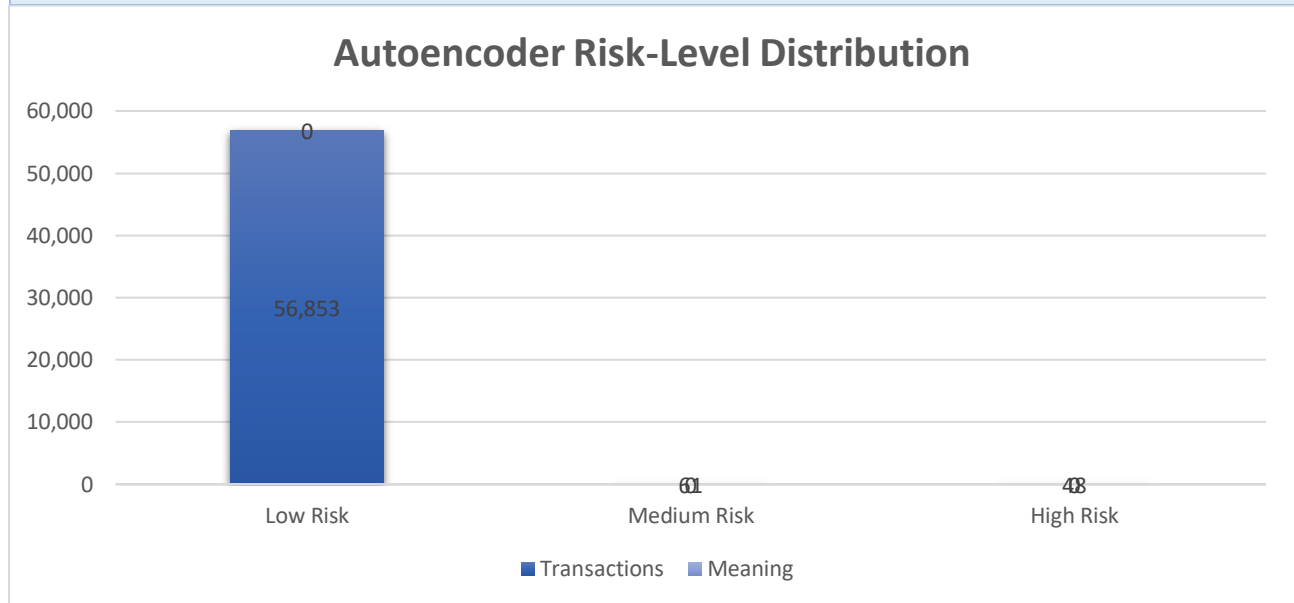
Recorded ROC-AUC values were 0.9539 for Isolation Forest and 0.9666 for the Autoencoder.



12. Fraud Alert System

The Autoencoder reconstruction error was converted into Low Risk, Medium Risk and High Risk categories. The High Risk category used an error greater than twice the selected threshold.

Risk Level	Transactions	Meaning
Low Risk	56,853	Normal reconstruction pattern
Medium Risk	61	Review recommended
High Risk	48	High-priority alert



The Autoencoder generated 109 anomaly alerts, including 48 High Risk alerts. At the selected threshold, 50 of 98 known fraud cases were detected.

13. Key Findings

- Fraud accounts for only 0.1727% of all transactions.
- Transaction amounts are highly right-skewed.
- Isolation Forest achieved 83.67% recall but generated 2,099 alerts.
- The Autoencoder achieved 45.87% precision, 51.02% recall, 48.31% F1-score, 96.66% ROC-AUC and 48.39% PR-AUC.
- The Autoencoder generated 109 alerts at the selected threshold, including 48 High Risk alerts.

14. Limitations

- V1–V28 are anonymized transformed variables and cannot be given direct business meanings.
- Historical data may not represent future fraud patterns.
- Anomaly detection can generate false positives and false negatives.
- Thresholds should be calibrated to business costs and investigation capacity before production use.

15. Future Scope

- Hyperparameter tuning and alternative anomaly-detection methods.
- Cost-based threshold optimization.
- Explainable AI for analyst-facing alerts.
- Real-time transaction scoring and alert delivery.
- Data-drift monitoring and periodic retraining.

16. Conclusion

This project demonstrates an end-to-end financial fraud detection workflow using exploratory analysis, class-imbalance-aware evaluation, Isolation Forest, an Autoencoder, and a risk-based alert system. The two methods show different operating trade-offs and provide a foundation for threshold calibration, model tuning, explainability and real-time monitoring.